Week 1Environment Setup Manual: The Waiting Problem

Concurrency & Distributed Systems — Lebanese University, Faculty of Engineering

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— Week 1 Lab

This section provides a complete PowerShell-first setup guide for configuring the Week 1 Lab environment on Windows.

The environment includes:

- Temurin JDK 25 (LTS)
- Visual Studio Code
- Java Extension Pack

Important — Administrative Privileges Required

All installation commands must be executed in PowerShell running as Administrator.

How to open PowerShell as Administrator:

1. Open the Start Menu
2. Type `PowerShell`
3. Right-click on **Windows PowerShell**
4. Select **Run as administrator**

If PowerShell is not running as Administrator, system-wide PATH and JDK installation will fail.

Step 0 — Verify or Install Winget

0.1 Check if winget is installed

```
winget --version
```

If a version number is displayed, proceed to Step 1.

0.2 If winget is NOT installed

Winget is part of the **App Installer** package in Windows 10/11.

Method 1 — Install via Microsoft Store:

- Open Microsoft Store
- Search for **App Installer**
- Install or Update it

Method 2 — Install manually (offline method)

Download the latest App Installer package from:

```
https://aka.ms/getwinget
```

After installation, close and reopen PowerShell, then verify:

```
winget --version
```

Step 1 — Install JDK 25 (Temurin LTS)

1.1 Install Temurin JDK 25

```
winget install -e --id EclipseAdoptium.Temurin.25.JDK --source winget
```

Step 2 — Configure JAVA_HOME and PATH

2.1 Locate the JDK installation directory

```
Get-ChildItem "C:\Program Files\Eclipse Adoptium\" -Directory |
Select-Object Name, FullName
```

Select the directory beginning with `jdk-25...`

2.2 Set JAVA_HOME and update system PATH

```
$jdk = "C:\Program Files\Eclipse Adoptium\jdk-25.0.x-hotspot"

[Environment]::SetEnvironmentVariable("JAVA_HOME", $jdk, "Machine")

$machinePath = [Environment]::GetEnvironmentVariable("Path","Machine")
$newMachinePath = "$jdk\bin;$machinePath"
[Environment]::SetEnvironmentVariable("Path", $newMachinePath, "Machine")
```

2.3 Refresh environment variables (no reboot required)

```
$env:Path = [Environment]::GetEnvironmentVariable("Path","Machine") +
";" +
[Environment]::GetEnvironmentVariable("Path","User")

$env:JAVA_HOME =
[Environment]::GetEnvironmentVariable("JAVA_HOME","Machine")
```

2.4 Verify installation

```
where.exe java
where.exe javac
java -version
javac -version
echo $env:JAVA_HOME
```

The first path shown must reference the Eclipse Adoptium directory.

Step 3 — Install Visual Studio Code

```
winget install -e --id Microsoft.VisualStudioCode --source winget
```

Verify installation:

```
code --version
```

If the command is not recognized, restart PowerShell.

Step 4 — Install Java Extension Pack

```
code --install-extension vscjava.vscode-java-pack
```

Optional explicit installations:

```
code --install-extension redhat.java
code --install-extension vscjava.vscode-maven
code --install-extension vscjava.vscode-gradle
```

Step 5 — Open and Run Week 1 Lab

5.1 Open the project

```
cd D:\courses\concerent\week1-lab
code .
```

5.2 Compile and run manually

```
cd D:\courses\concerent\week1-lab\lab1
javac *.java
java SingleThreadedServer
```

In a second terminal:

```
java LoadTestClient
```

Step 6 — Sanity Test (Hello World)

```
cd D:\courses\concerent\week1-lab\lab1

@'
public class Hello {
  public static void main(String[] args) {
    System.out.println("Hello Java 25!");
  }
}
'@ | Set-Content .\Hello.java

javac Hello.java
java Hello
```

Successful output confirms correct configuration.

A correct environment setup eliminates 90% of concurrency debugging problems.

